

PRD: Fuzhounese Dictionary — a community dictionary of 福州話

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How to use this document. This is a *right-sized* PRD for the first buildable release of a community dictionary. It is a real platform — a web frontend, a database, sign-in, file storage, and a moderation step — which is a larger genre than a single-file app, so the discipline matters more, not less: v1 is one coherent release, every requirement below is numbered so it can be pointed at (“implement FR-11,” “FR-16 is broken”), everything speculative is a non-goal or an open question, and the intended reader is a developer who knows nothing about the project.

1. Overview

The Fuzhounese Dictionary is a web app where anyone can look up a Fuzhounese word — by Chinese characters, by romanization, or by its English or Chinese meaning — and where signed-in contributors can add new words. Every submission goes into a moderation queue and appears publicly only after an editor approves it. The dictionary data is openly licensed so the community can trust and reuse it.

Problem. Fuzhounese (福州話 · Foochow · Eastern Min) has millions of speakers and a large overseas diaspora, but it is thinly documented online. What exists is scattered, inconsistent in romanization, and almost never lets ordinary speakers add the words they know. A heritage speaker who wants to check a spelling, find the characters for a word, or record a word before it is forgotten has nowhere central to go — and no way to contribute what they know back.

Who it’s for. Two primary audiences: **diaspora heritage speakers** who speak some Fuzhounese and want to look words up, confirm spellings/characters, and preserve vocabulary; and **language learners** who need pronunciations, meanings, and example usage. Linguists and researchers benefit but are not the group the product is designed around.

2. Goals

G1 — A visitor can find a word by characters, romanization, or English/Chinese meaning in a few seconds, with tone marks optional.

G2 — A signed-in contributor can add a fully-formed word entry in a couple of minutes, and it reaches an editor for review.

G3 — An editor can approve or reject a submission in one action, and approved words appear on the site immediately.

G4 — An entry can carry everything a real dictionary needs: characters, free-form romanization, IPA, audio, and one or more senses (each with an English definition and an optional Chinese gloss), plus examples and notes.

G5 — Contributors are credited, and the dictionary's content is released under an open license (CC BY-SA).

G6 — The whole thing runs on the free tiers of Supabase and Vercel.

3. Non-goals

Stating what we are *not* building is the highest-leverage part of this PRD. These fence off the features most likely to balloon the build.

NG1 — No enforced romanization standard. Contributors write romanization however they know it; the app does not validate, convert, or auto-tone Bàng-uâ-cê. (Search is tone-insensitive, so this still works — see FR-2.)

NG2 — No public corrections / edit-suggestion workflow in v1. Fixing an existing entry is an editor action; users cannot yet propose an edit to a specific word.

NG3 — No anonymous submissions. Contributing requires signing in.

NG4 — No native mobile app. Responsive web is the platform.

NG5 — No discussion, comments, forums, upvoting, or social feed.

NG6 — No full-sentence or machine translation. This is a word dictionary.

NG7 — No offline mode.

NG8 — No multi-tier editorial roles or permissions in v1. There is one owner-editor (see Q-4).

4. User stories

Visitor (no account)

US-1 — As a visitor, I can search a word by Chinese characters, romanization, or English/Chinese meaning, so I can find it however I happen to know it.

US-2 — As a visitor, I can open a word and see its characters, pronunciations (romanization, IPA, audio), meanings, and examples, so I understand it and can say it.

US-3 — As a visitor, I can browse all words alphabetically, so I can explore without a specific query.

US-4 — As a visitor, when my search finds nothing, I'm invited to contribute that word, so gaps get filled.

Contributor (signed in)

US-5 — As a visitor, I can sign in with Google, so I can contribute.

US-6 — As a contributor, I can submit a new word with all its fields, so I can grow the dictionary.

US-7 — As a contributor, I can attach an audio pronunciation (upload a file or paste a link), so others can hear the word.

US-8 — As a contributor, I can add more than one sense to a word, so a word with several meanings is captured fully.

US-9 — As a contributor, I can see which of my submissions are pending, approved, or rejected, so I know the status of my work.

US-10 — As a contributor, I'm credited on entries I add, so my contribution is visible.

Editor

US-11 — As an editor, I can see all pending submissions in one queue, so I can review them.

US-12 — As an editor, I can approve or reject a submission with an optional note, so quality stays high and the contributor learns why.

US-13 — As an editor, I can edit any entry, so I can fix mistakes without a separate corrections system.

US-14 — As an editor, I can bulk-import my own pre-approved word list, so I can seed a few hundred words quickly.

5. Functional requirements

Numbered, explicit, one behavior each. If a requirement needs the word “and” twice, it's split.

Search & display

FR-1 — The home page has a single search box. Submitting a query returns approved entries whose characters, romanization, IPA, English definition, or Chinese gloss match the query, ranked by relevance.

FR-2 — Search ignores tone diacritics and letter case: `sioh` matches `siōh`, and `hok` matches `Hók`.

FR-3 — Each search result shows the characters (if any), the romanization, the part of speech, and the first sense's short gloss, and links to the full entry.

FR-4 — An entry page shows: characters, romanization(s), IPA, an audio player (if audio exists), and each sense with its part of speech, English definition, Chinese gloss, example sentence, example translation, and notes.

FR-5 — When an entry has no Chinese characters (many colloquial Min morphemes have no standard hàn-cê), the character field is simply omitted; the entry is still valid and fully displayable from its romanization alone.

FR-6 — A browse page lists all approved entries alphabetically by romanization, paginated.

FR-7 — When a search returns zero results, the page shows a “Know this word? Add it” call to action that pre-fills the submission form with the query. If the visitor is not signed in, it prompts sign-in first.

Accounts & submission

FR-8 — A user can sign in with Google (OAuth, via Supabase Auth). There are no passwords.

FR-9 — Only signed-in users can open or submit the “Add a word” form. Signed-out users see a sign-in prompt in its place.

FR-10 — The submission form accepts, per entry: characters (optional), romanization (optional, free-form), IPA (optional), and audio (optional — a file upload or a URL); plus one or more *senses*. Each sense has: part of speech (optional), English definition (**required**), Chinese gloss (optional), example sentence (optional), example translation (optional), and notes (optional).

FR-11 — A submission is valid only if it has at least one identifying form (characters or romanization) *and* at least one sense with an English definition. An invalid submission shows an inline error and saves nothing.

FR-12 — Audio upload accepts common audio formats up to a fixed size limit (target 5 MB); a pasted audio URL is stored as-is. Uploaded files go to a Supabase Storage bucket.

FR-13 — On submit, the entry is saved with status `pending` and stamped with the contributor’s user id, and the contributor sees a confirmation.

FR-14 — A contributor has a “My submissions” view listing their entries with each one’s status (pending / approved / rejected) and any editor note.

Moderation

FR-15 — Editors reach a moderation queue at a protected route, listing pending submissions oldest-first with every field shown and any attached audio playable inline.

FR-16 — An editor can **approve** a submission (status → `approved` ; it appears publicly at once) or **reject** it (status → `rejected` ; hidden but kept on record), with an optional review note shown to the contributor.

FR-17 — An editor can edit any entry’s fields at any time, whatever its status.

FR-18 — An approved entry shows a contributor credit (display name), except for entries the owner bulk-imported.

FR-19 — Editors can bulk-import entries from a CSV as status `approved` , bypassing the queue.

Content & platform

FR-20 — Every page footer states the content license (CC BY-SA) and links to it.

FR-21 — The layout is usable at 375px wide (phone) and 1280px wide (laptop) without horizontal scrolling.

FR-22 — Row-Level Security guarantees the public/anonymous role can read only `approved` entries — never pending or rejected ones — and that approve, reject, and edit happen only through a privileged server path.

6. Design considerations

- Reading-first and quiet. A single centered column, roughly 720px max on reading pages; search results may run wider. The visual hierarchy inside an entry is always character → pronunciation → meaning.
- Characters are shown large in a serif CJK face; romanization in an accent color; meanings in plain body text.
- Tone diacritics must render correctly, including stacked combining marks (ǖ, è, ā, ě). Use a Unicode-complete font stack and verify these glyphs on a real device.
- Missing data degrades gracefully: absent characters, audio, examples, or glosses simply disappear rather than leaving empty labels.
- Respect the OS light/dark preference.
- Audio is a plain native player. On mobile, the search box is the hero element.

7. Technical considerations

TC-1 — Next.js (App Router, TypeScript) frontend; Supabase (hosted Postgres + Auth + Storage) backend; deploy on Vercel. Targets the free tier of each.

TC-2 — Senses are a child table, not a JSON blob, so search and display treat every sense uniformly and a word can have many. Data shape (simplified):

```
entries { id, headword, hanzi?, romanization?, ipa?, audio_url?,
          notes?, status, contributor_id, created_at, reviewed_at }
senses  { id, entry_id, part_of_speech?, definition_en,
          gloss_zh?, example?, example_gloss?, sort }
profiles { id (= auth uid), display_name, is_editor }
```

TC-3 — Search is a Postgres function combining full-text (`tsvector`), trigram (`pg_trgm`), and `unaccent` , so both tone-insensitive and partial matches work; the client calls it via RPC.

TC-4 — Auth is Supabase Auth with Google OAuth. A `profiles` row is created on first sign-in. An `is_editor` boolean, set manually, unlocks the moderation queue and edit actions.

TC-5 — Row-Level Security: anyone may `SELECT` only `status = 'approved'` ; an authenticated user may `INSERT` an entry as `pending` with `contributor_id` equal to their own uid; approve / reject / edit are restricted to the editor role (or performed through a service-role server action).

TC-6 — Audio lives in a Supabase Storage bucket: authenticated users may upload to their own folder; files are publicly readable; the entry stores the public URL. Size and type are enforced on the client and by a bucket policy.

TC-7 — No secret keys ever reach the browser. The service-role key is used only in server code paths.

8. Success metrics

SM-1 (*primary — coverage*) — The dictionary grows from its initial seed (a few hundred words) toward 1,000+ approved entries, and the live count is visible on the home page.

SM-2 (*primary — findability*) — For a fixed set of words known to be in the dictionary, searching *any* of their forms — characters, romanization with or without tone marks, or English — returns the entry in the top results.

SM-3 (*secondary — community*) — A measurable share of newly approved words come from community submissions, not only the owner’s bulk import.

SM-4 (*secondary — usage*) — Visitors return and look words up: a basic count of searches per week and entry views trends upward after launch.

9. Open questions

Q-1 — Which variety is the reference? “Fuzhounese” spans the city dialect and county varieties (Changle, Lianjiang, Fuqing/Hokchia, Gutian, Matsu...), and much of the overseas diaspora speaks specific county forms. Do we treat the Fuzhou-city form as the headword and mark regional variants, or accept all and tag each entry with its variety? (*Assume: accept all, add an optional “variety / region” tag — confirm.*)

Q-2 — Duplicates. With no corrections flow (NG2), what happens when a contributor submits a word that already exists — reject as duplicate, allow a second entry, or route to the editor to merge? (*Assume: the editor decides at review, and the queue shows a “possible duplicate” hint.*)

Q-3 — Chinese gloss. Is it specifically Mandarin, and is it ever required? (*Assume: an optional Mandarin gloss.*)

Q-4 — Editorial team. v1 assumes a single owner-editor. When do we need multiple editors with roles, and an audit of who approved what? (*Out of scope for v1; flag when a second reviewer is needed.*)

Q-5 — Spam & abuse. Beyond requiring sign-in, do we need rate limits, a captcha, or a trusted-contributor tier as volume grows? (*Assume: sign-in + moderation are enough for v1.*)

Q-6 — Name & domain. “Fuzhounese Dictionary” is a working title. Is there a name you want, in English and/or 福州話?

Q-7 — Audio review load. Audio makes moderation heavier — the editor has to listen. Is audio reviewed inline in v1 (current assumption), or added only by editors if community audio proves noisy?

Notes on why this document is shaped the way it is

It is scoped to one buildable release. This is a platform — frontend, database, auth, storage, and a moderation step — which the template rightly calls a different genre from a single-file app. The response is not to write a bigger, vaguer document but a tighter one: v1 is a single coherent release, and everything past it lives in Non-goals or Open questions.

Non-goals do the heavy lifting. With any capable builder, the failure mode is too much, not too little — a corrections workflow, multi-editor roles, and romanization validation are exactly the features that would triple the build. Section 3 is the fence.

Every requirement is numbered and testable. “Search should be forgiving” is not a requirement; FR-2 is, because you can write down an input (`sioh`) and its expected output (`siǒh`) — which is exactly what SM-2 then checks.

Ambiguity is named, not hidden. Section 9 says out loud what hasn’t been decided — above all, which variety of Fuzhounese is the reference. A builder who hits an undecided question will decide it silently and confidently; this document hands those decisions back to you.

The workflow this feeds. PRD → task list → implement one requirement at a time, reviewing each. The PRD is the durable artifact; the task list is generated from it and thrown away.